

Statistics For Textile Engineering

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Preface

This eBook is developed for the students of bachelor students of Texttile Engineering.

1 Introduction to Statistics

The field of **statistics** deals with the collection, presentation, analysis, and use of data to make decisions, solve problems, and design products and processes. In simple terms, **statistics is the science of data**.

Because many aspects of engineering practice involve working with data, obviously knowledge of statistics is just as important to an engineer as are the other engineering sciences.

Specifically, statistical techniques can be powerful aids in designing new products and systems, improving existing designs, and designing, developing, and improving production processes.

Statistical methods are used to help us describe and understand **variability**. By variability, we mean that successive observations of a system or phenomenon do not produce the same result. We all encounter variability in our everyday lives, and **statistical thinking** can give us a useful way to incorporate this variability into our decision-making processes (Montgomery and Runger 2014).

1.1 Types of Data

- **Quantitative:** Data that represents numerical values.

Example: Heights of people, temperatures, test scores.

Subtypes:

- **Discrete Data:** Countable values (e.g., number of students in a class).
- **Continuous Data:** Measurable values that can take any value within a range (e.g., weight, time).

- **Categorical:** Data that represents categories or labels.

Example: Colors of cars, types of animals, survey responses (e.g., yes/no).

Subtypes:

- **Nominal Data:** Categories without a natural order (e.g., gender, blood type).
- **Ordinal Data:** Categories with a meaningful order (e.g., rankings, education levels).

1.2 Source of data

When the data are all of the observations in the population. This results in a **census**. However, in the engineering environment, the data are almost always a **sample** that has been selected from the population. Three basic methods of collecting data are

(i) A **retrospective study** using historical data (ii) An **observational study** (iii) A **designed experiment**

An effective data-collection procedure can greatly simplify the analysis and lead to improved understanding of the population or process that is being studied. We now consider some examples of these data-collection methods. (For detail, see Montgomery and Runger (2014) , **Chapter 1**)

1.3 Population and Sample

- **Population:** A **population** is the entire collection of elements or outcomes about which information is required.

The total number of elements in the population is called the **population size**, denoted by N .

- **Sample:** A **sample** is a subset of a population, comprising the elements or outcomes that are actually observed. A good sample must be *representative* of its population.

The number of elements in the sample is called the **sample size**, denoted by n .

- A numerical summary of a sample is called a **statistic**.
- A numerical summary of a population is called a **parameter**.
- **Statistics** are often used to *estimate* **parameters**.

1.4 Random Sample

A simple random sample of size n is a sample chosen by a method in which each collection of n elements selected from population is equally likely to contain the sample, just as in a lottery.

1.5 Branch of Statistics: Descriptive and Inferential Statistics

The study of statistics is divided into two parts- **Descriptive statistics** and **Inferential statistics**.

Descriptive statistics is the branch of statistics that involves the organization, summarization, and display of data.

Inferential statistics is the branch of statistics that involves using a sample to draw conclusions about a population.

A basic tool in the study of inferential statistics is **probability** and **probability distributions**.

The goal of inferential statistics is to draw conclusions from a sample and generalize them to the population. The most common methodologies used are hypothesis tests, Analysis of variance, etc.

We will begin with **descriptive statistics**.

1.6 Descriptive Statistics

1.6.1 Measures of Central Tendency

Central tendency refers to a most typical or most representative or central value of a data set. It summarize a data numerically with a single measure. The most common **measures of central tendency** are the **mean**(**arithmetic average**), **median** and **mode**.

1.6.1.1 The mean

Consider a sample data of size n of variable X with observations $x_1, x_2, \dots, x_n$. Then the mean/sample mean of X is:

$$\bar{X} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum x}{n} \quad (1.1)$$

On the other hand if we have population data of size N then the population mean of X is:

$$\mu = \frac{\sum_{i=1}^N x_i}{N} \quad (1.2)$$

Example:

Note:

- Sum of the deviation from mean is zero that is $\sum_{i=1}^n (x_i - \bar{x}) = 0$.
- For any real number $a \neq \bar{x}$, $\sum_{i=1}^n (x_i - \bar{x})^2 < \sum_{i=1}^n (x_i - a)^2$.
- Mean is affected by outlier/extreme value.

1.6.1.2 The median

1.6.1.3 The mode

1.6.2 The quantiles (Percentiles, quartiles, deciles)

The quantiles are cut points which divide a data set into particular equal portions. For example:

- Percentiles divides the data into approximately 100 equal parts.
- Quartiles divides the data into approximately 4 equal parts.
- Deciles divides the data into approximately 10 equal parts.

Quartiles and Deciles are special case of Percentiles.

Formula to compute Percentiles

The j^{th} percentile is computed as follows:

$$P_j = \left(\frac{n+1}{100} * j \right)^{th} \text{ value} ; j = 1, 2, \dots, 99 \quad (1.3)$$

Note that

- First/lower quartile, $Q_1 = P_{25}$.
- Second quartile/median, $Q_2 = P_{50}$.
- Third/upper quartile, $Q_3 = P_{75}$.

Problem Compute the 25th and 60th percentile from the following data:

CPU time of 9 jobs (in seconds): 59, 46, 37, 42, 30, 55, 56, 36, and 82.

Solution: Here, $n = 9$.

Sorted data: 30, 36, 37, 42, 46, 55, 56, 59, and 82.

So, $P_{25} = (25 * \frac{9+1}{100})^{th} \text{ value} = 2.5^{th} \text{ value}$
 $= 2^{nd} \text{ value} + 0.5(3^{rd} - 2^{nd}) = 36 + 0.5(37 - 36) = 36.5$.

Interpretation: $P_{25} = 36.5$ implies that approximately 25% of the total observations lie below or equal to 36.5.

Similarly, $P_{60} = (60 * \frac{9+1}{100})^{th} \text{ value} = 6^{th} \text{ value} = 55$.

Interpretation: $P_{60} = 55$ implies that approximately 60% of the total observations lie below or equal to 55.

Problem : Here is the monthly starting salary (\$) of 12 graduates:

3450, 3550, 3650, 3480, 3355, 3310, 3490, 3730, 3540, 3925, 3520, 3480

Compute Q1 and Q3 of the above data (*Will be solved in class*).

1.6.3 Measures of Variation

Variability in data means lack of uniformity. It is also referred to as spread, scatter, or dispersion. We turn now to a discussion of some commonly used measures of variability.

1.6.3.1 Range

Range, $R = \text{Maximum value} - \text{Minimum value}$

- Heavily influenced by extreme values

1.6.3.2 Inter-quartile-range

Inter-quartile-range, $IQR = Q_3 - Q_1$

- It exhibits the variability in the middle 50% of the observations.
- The interquartile range is less sensitive to the extreme values in the sample than is the ordinary sample range.

1.6.3.3 Variance and Standard deviation

- Sample variance (denoted by English Letter s^2),

$$s^2 = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2}{n - 1} = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$
$$\therefore s^2 = \frac{\sum x_i^2 - n(\bar{x})^2}{n - 1} \quad (1.4)$$

and Sample SD,

$$s = \sqrt{s^2}$$

- Population variance (denoted by Greek Letter σ^2),

$$\sigma^2 = \frac{(x_1 - \mu)^2 + (x_2 - \mu)^2 + \dots + (x_N - \mu)^2}{N} = \frac{\sum (x_i - \mu)^2}{N} = \frac{\sum x_i^2 - N(\mu)^2}{N}$$
$$\therefore \sigma^2 = \frac{\sum x^2}{N} - \mu^2 \quad (1.5)$$

and Population SD is

$$\sigma = \sqrt{\sigma^2}$$

Note: In modern statistics and quality management, **consistency is the hallmark of quality**. While the **Mean** tells you if you are hitting your target, the **Variation (Standard Deviation)** tells you how reliable you are. High variation is often referred to as the “enemy” of quality because it creates unpredictability.

Problem : Ten-count tests made on certain yarn have shown following results.

22.8, 23.2, 22.9, 22.6, 23.4, 23.0, 23.1, 23.0, 22.9, 23.0

Compute sample mean, sample variance and sample standard deviation.

Solution: Suppose X be the count test.

Here, $n = 10$; $\sum x = 229.9$,

$$\sum x^2 = x_1^2 + x_2^2 + \dots + x_n^2 = 22.8^2 + 23.2^2 + \dots + 23.0^2 = 5285.83$$

So, the **sample mean** $\bar{x} = \frac{\sum x}{n} = \frac{229.9}{10} = 22.99$.

Sample variance, $s^2 = \frac{\sum x^2 - n \cdot (\bar{x})^2}{n-1} = \frac{5285.83 - 10 \cdot (22.99^2)}{10-1} = 0.04766667$ and

Sample SD, $s = \sqrt{s^2} = \sqrt{0.04766667} = 0.218327 \approx 0.2183$.

1.6.3.4 Coefficient of variation

$$CV = \frac{SD}{Mean} \quad (1.6)$$

- CV is unit free.
- A lower CV indicates that the variation is small compared to the size of the mean, which refers **greater consistency** or **reliability**.
- In general industry Standards for “Good Consistency”, **CV should be less than 10%** and $CV \geq 30\%$ indicates “Poor consistency” (Belle 2002) .

Problem: **Compute** coefficient of variation of yarn-count data mentioned in Problem.....

Problem Five strength tests each carried out on the two fabrics have shown following results:

A: 15.2, 15.5, 15.0, 15.4, 15.6

B: 15.0, 14.8, 13.6, 15.5, 15.2

Find CV% for both and **decide** which fabric is more consistent.

Solution:

Problem Following data represents number of end breaks recorded on two ring frames at ten occasions.

R/F-I: 12, 10, 8, 15, 9, 11, 14, 15, 8, 10

R/F-II: 10, 11, 12, 9, 12, 9, 11, 12, 11, 10

Find which ring frame is more consistent in terms of end breaks.

Solution:

i Exercise 1.1

1) An article describes an experiment to test the yield strength of circular tubes with caps welded to the ends. The first 8 yields (in kN) are: 104, 108, 126, 126, 128, 128, 140, 156
Compute the sample mean, the standard deviation, and the coefficient of variation of the yield strength.

2) The “cold start ignition time” of an automobile engine is being investigated by a gasoline manufacturer. The following times (in seconds) were obtained for a test vehicle: 1.75, 1.92, 2.62, 2.35, 3.09, 3.15, 2.53, 1.91.

Calculate the sample mean, sample standard deviation and coefficient of variation of cold start ignition time.

3) The monthly starting salary (\$) of 12 graduates are as follows:

3450 ,3550 ,3650 ,3480 ,3355, 3310 ,3490 ,3730, 3540 ,3925, 3520 ,3480

(i) **Compute** mean, median, lower and upper quartiles of salary data. Also compute IQR and standard deviation

(ii) To be in top 15% earners what should be the starting salary of a graduate?

4) The lengths of power failures, in minutes, are recorded as follows: 18, 15, 90, 78, 69, 98, 102, 83, 55, 28.

(i) **Compute** sample mean and standard deviation.

(ii) **Compute** sample median and IQR.

(iii) To be in the bottom 10% of power failures in your residence then what should be the cutoff value of power failure lengths?

5) Following are the results of count tests made on the production of two shifts:

Shift I: 11.7, 12.2, 12.0, 11.6, 10.9, 13.0, 11.0, 12.8, 11.5, 12.0

Shift II: 12.2, 11.6, 12.0, 12.1, 12.0, 12.3, 11.8, 11.6, 11.8, 12.0

Using CV% **decide** which production is more consistent.

1.7 Data visualization

Before diving into advanced analysis we should have a look at the data. Because ” A picture is worth a thousand words”. Often a summary of a collection of data via a graphical display can provide insight regarding the system from which the data were taken.

1.7.1 Data viz. for Qualitative/categorical data

- **Bar-chart**- used to visualise frequency of each label of categorical variable with 2D bar.

- **Pie-chart**- used to visualise frequency/ percentage of each label of categorical variable in a circle.
- **Stacked/Clustered Bar chart**- used to show the relation between two categorical variables in side-by-side column/ stacked-column.

For detail, visit [Summarizing Categorical Data](#).

1.7.2 Data viz. for Quantitative data:

- Histogram
- Boxplot
- Scatter plot

1.7.2.1 Histogram

The **histogram** is a visual display of the frequency distribution. **The histogram** provides a visual impression of the **shape** of the distribution of the measurements and information about the **central tendency** and scatter or **dispersion** in the data.

To construct a frequency distribution, we must divide the range of the data into intervals, which are usually called **class intervals**, **cells**, or **bins (number of classes)** , and count how many observations fall into each **bin**.

- A **frequency histogram** consists of columns, one for each **bin**, whose height is determined by the **number** of observations in the **bin**. [Frequency distribution]
- A **relative frequency histogram** has the same shape but a different vertical scale. Its column heights represent the **proportion** of all data that appeared in each bin. [Relative frequency distribution]

Example 1.10: The following data are the joint temperatures of the O-rings (°F) for each test firing or actual launch of the space shuttle rocket motor (from *Presidential Commission on the Space Shuttle Challenger Accident*, Vol. 1, pp. 129–131):

67, 40, 58, 76, 58, 70, 72, 67, 75, 70, 57, 83, 53, 45, 70, 81, 78, 76, 67, 73, 61, 52, 31, 67, 79, 75, 69, 84, 68, 80

To construct frequency distribution and others follow the steps:

Step-1: Find the **number bins**, $k = \sqrt{n} = \sqrt{30} = 5.48 \approx 6$ (round to nearest integer).

Step-2: Find the range of the data, $R = \text{Maximum} - \text{Minimum} = 84 - 31 = 53$

Step-3: Determine **bin width**, $w = \frac{R}{k} = \frac{53}{6} = 8.83 \approx 10$

Step-4: Define the **bins** in *exclusive* method, starting from a suitable data value close to lowest value, for example **[30, 40)**, **[40, 50)**, **and so on** until we have the highest data value.

Step-5: Now count the observation fall in each bin, using tally.

Table 1.1: Frequency, Relative frequency, Cumulative frequency Distribution for the O-rings temperature data (n=30)

Temperatures (in ° F)	Tally	Frequency (f)	Relative frequency (rf)	Cumulative frequency (cf)
[30,40)		1	0.03	1
[40,50)		2	0.07	3
[50,60)		5	0.17	8
[60,70)		7	0.23	15
[70,80)		11	0.37	26
[80,90)		4	0.13	30
Total		n=30	1.00	

The histogram from the previous example is shown in Figure 1.1:

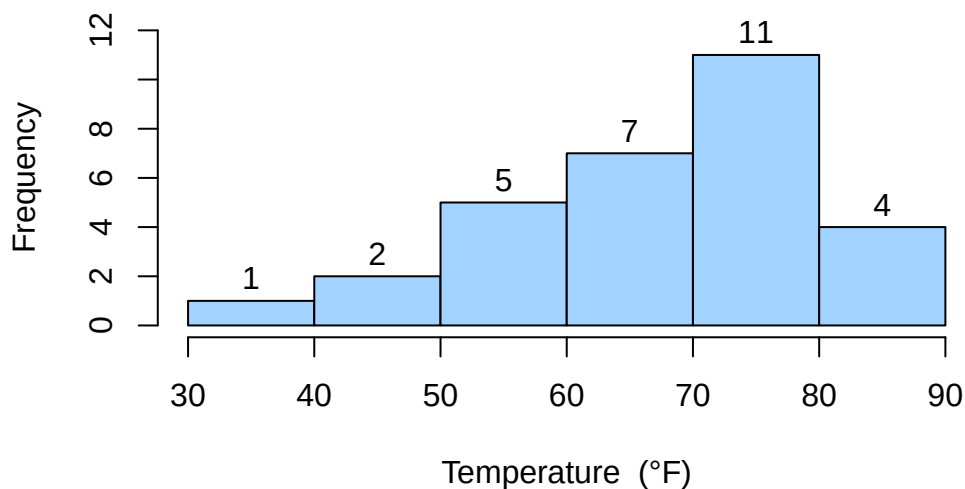


Figure 1.1: Frequency histogram of temperature (°F)

Histogram and shape of the distribution

Departure from symmetry is often referred as **skewness**. When the sample size is large, the histogram can provide a reasonably reliable indicator of the general **shape** of the distribution or population of measurements from which the sample was drawn. Here are some typical examples of **skewness**.

NOTE: A **symmetric histogram** often represents a stable, predictable process where data clusters evenly around the mean, indicating that variation is likely due to common causes rather than specific errors.

1.7.2.2 Box-plot

The **box plot** is a graphical presentation that simultaneously describes several important features of a data set, such as **center**, **spread**, **departure from symmetry**, and identification of unusual observations or **outliers**.

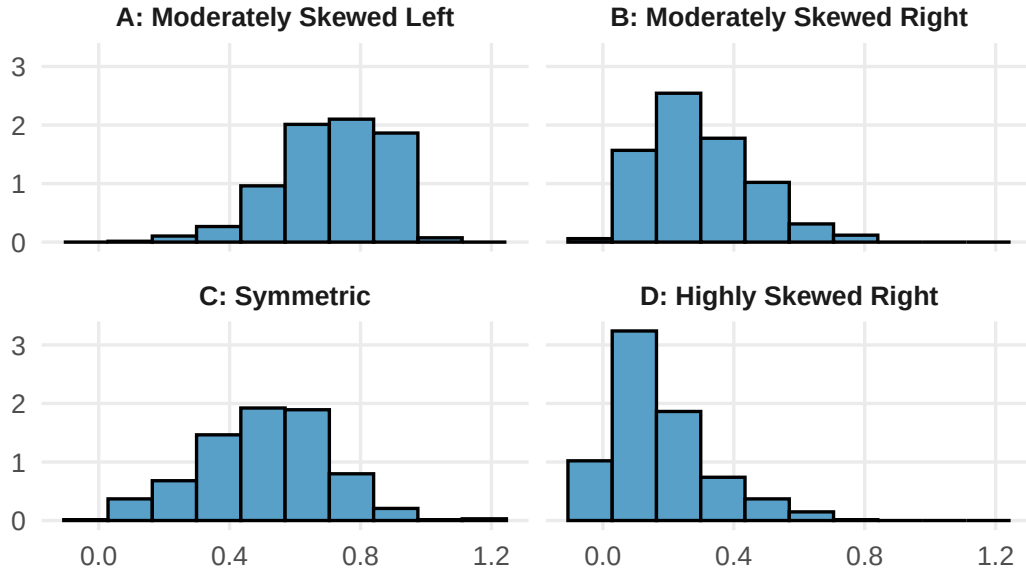


Figure 1.2: Histograms Showing Differing Levels of Skewness

To construct a box-plot we have

- to sort data (smallest to largest)
- to compute Q_1, Q_2, Q_3, IQR
- to determine the **lower fence**, $lf = Q_1 - 1.5(IQR)$
- to determine the **upper fence**, $uf = Q_3 + 1.5(IQR)$

Then we draw a diagram like the following which is called a box plot (Montgomery and Runger 2014, 239).

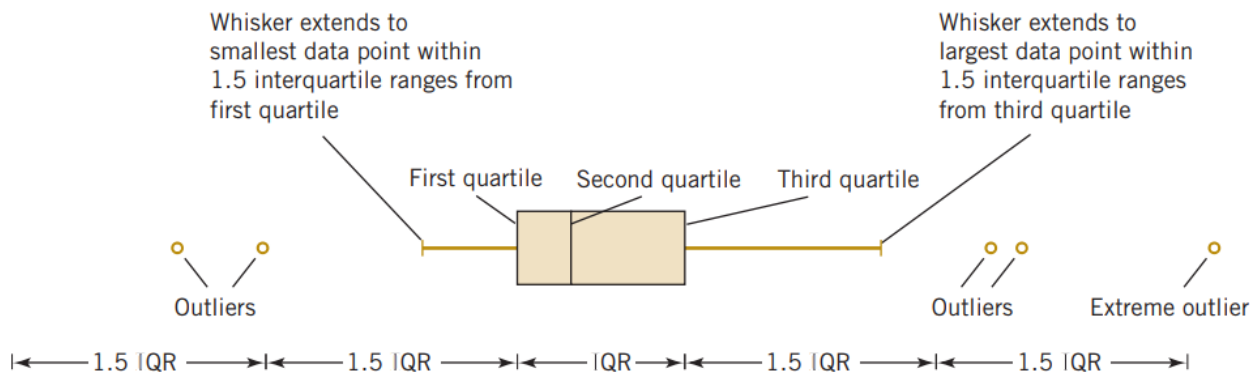


Figure 1.3: Description of a box plot

Illustration with an example

The data provides a the fuel consumption in Gallons per Kilometer (GpK) for a group of 17 medium-sized mobile home units.

30, 34, 34, 34, 34, 35, 35, 35, 35, 35, 35, 37, 37, 38, 40, 42, 42

- (a) **Construct** a box-plot.
- (b) **Comment** about shape of the data (departure from symmetry)
- (c) **Show** outlier (if any).

Solution:

(a) Box-plot

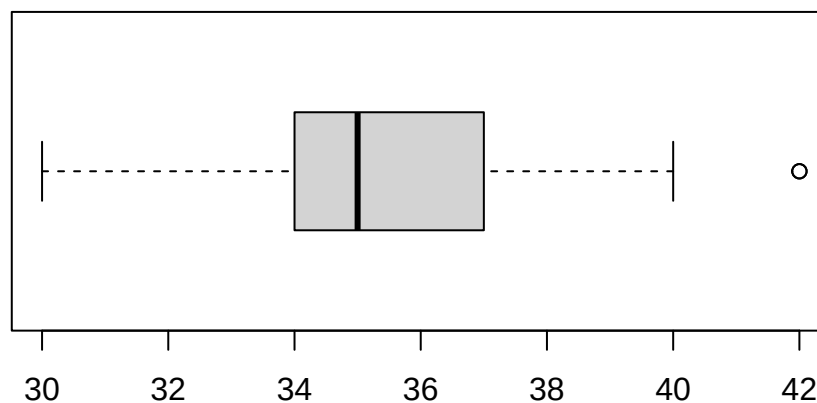


Figure 1.4: Box-plot of fuel consumption (in GpK)

- (b) Approximately symmetric.
- (c) There is an outlier in the upper tail (42).

Parallel/comparative box-plot

Box plots are very useful in graphical comparisons among data sets because they have high visual impact and are easy to understand.

Example: Two catalysts are being analyzed to determine how they affect the mean yield of a chemical process. Specifically, catalyst 1 is currently used; but catalyst 2 is acceptable. Because catalyst 2 is cheaper, it should be adopted, if it does not change the process yield. A test is run in the pilot plant and results in the data shown below (Montgomery and Runger 2014, 384).

Table 1.2: Performance Values for Catalyst 1 and Catalyst 2

Observation Number	1	2	3	4	5	6	7	8
Catalyst 1	91.50	94.18	92.18	95.39	91.79	89.07	94.72	89.21
Catalyst 2	89.19	90.95	90.46	93.21	97.19	97.04	91.07	92.75

Figure 1.5 presents a comparative box plot of the data from the two samples.

Though the **median** yields are same for both catalysts but Catalyst 2 produces higher variability in yield. However overall maximum yield is obtained from Catalyst 2. So based on this visual inspection we can use Catalyst 2 (it is also cheaper!).

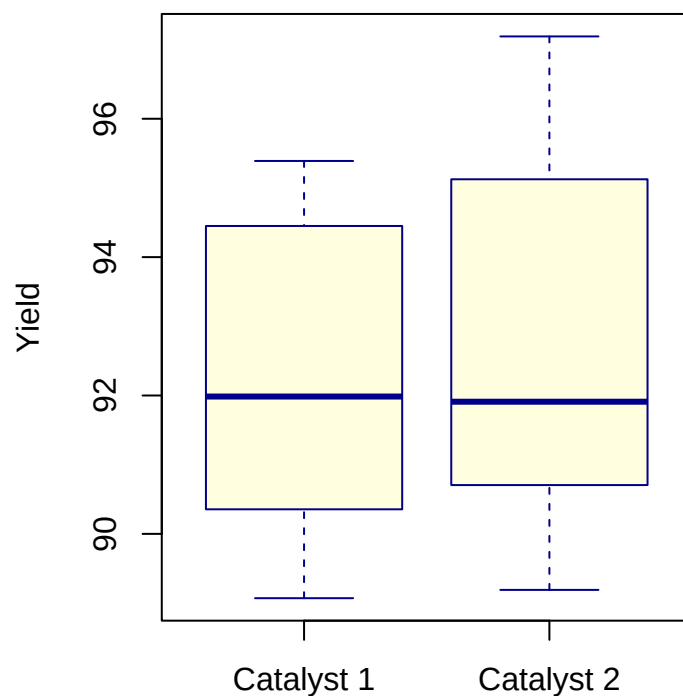


Figure 1.5: Comparison of Catalyst Performance: Catalyst 1 vs. Catalyst 2

NOTE: To know more about box-plot please visit [Minitab® Support](#).

i Exercise 1.2

1) The following sample data presents the thickness ($\text{\AA}$) of a metal layer on 20 silicon wafers resulting from a chemical vapor deposition (CVD) process. Scientists believe that the thickness is usually normally distributed having a bell-shaped distribution for a smooth process. **Construct** a frequency histogram of metal thickness data. **Conclude** whether the process is smooth or not.

468, 459, 450, 453, 473, 454, 458, 438, 447, 463,
445, 466, 456, 434, 471, 437, 459, 445, 454, 423

2) The shear strengths of 20 spot welds in a titanium alloy are as follows:

5408 5431 5475 5442 5376 5388 5459 5422 5416 5435
5420 5429 5401 5446 5487 5416 5382 5357 5388 5457

Construct a frequency histogram of shear strength data. **Conclude** whether the shear strength is approximately symmetric or not.

3) The data provides the fuel consumption for a group of 17 medium-sized mobile home units.

30, 34, 34, 34, 34, 35, 35, 35, 35, 35, 35, 37, 37, 38, 40, 42, 42.

i) **Construct** a boxplot and show outlier(s) if any. Does the plot appear to be symmetric?

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ii) **Find** the 90th percentile. Is it an outlier?

4) The cylindrical compressive strength (in MPa) was measured for 11 beams. The results

2 Probability and Random Variables

Our life is full of uncertainty. To take valid decision in the face of uncertainty we have to study **probability**. For examples we often think of the following phenomenon:

- What is the chance that a newly bought smartphone malfunction before the warranty period?
- What is the probability that 2 or more defective items in found in a lot?
- How likely that no customer arrives in next five-minute period in a superstore? etc.

2.1 Basic elements of probability

- **Random experiment:** Which has 2 or more outcomes and cannot be predicted with certainty. **Example:** Tossing a coin, rolling a dice, testing the strength of fibers etc.
- **Sample space:** The set of all possible outcomes of a random experiment, often denoted by Greek uppercase letter Omega (Ω). **Example** The sample space of rolling a six-sided dice is: $\Omega = \{1, 2, 3, 4, 5, 6\}$.
- **Event:** An event (A) is a subset of the sample space ($A \subseteq \Omega$).
- **Complement event:** The complement of an event A consists of all the outcomes in the sample space Ω that are not in A , often denoted by A^c, A' or $\bar{A}$. In set theory, $A' = \Omega - A$.
- **Mutually exclusive/disjoint events:** A sequence of events $A_1, A_2, A_3, \dots$ are said to be mutually exclusive events if they cannot happen together.

Mathematically,

$$A_i \cap A_j = \emptyset \quad \text{for } i \neq j$$

- **Mutually exhaustive events:** In probability, a set of events is called collectively exhaustive if their union covers the entire sample space. In other words, at least one of these events must happen whenever the experiment is performed. If $E_1, E_2, E_3, \dots, E_n$ are events in a sample space Ω , they are exhaustive if:

$$E_1 \cup E_2 \cup E_3 = \Omega$$

2.2 Probability Measure

A probability measure(P) is a function that assigns a probability to each event defined in Ω satisfies the following axioms:

- (i) $0 \leq P(A) \leq 1$
- (ii) For any countable sequence of disjoint events $A_1, A_2, A_3, \dots$,

$$P(A_1 \cup A_2 \cup A_3 \cup \dots) = P(A_1) + P(A_2) + P(A_3) + \dots$$

- (iii) $P(\Omega) = 1$

2.3 Discrete Uniform law of probability

If an experiment can result in any one of N different equally likely outcomes, and if exactly n of these outcomes correspond to event A , then the probability of event A is

$$P(A) = \frac{n}{N}$$

For example, if a box contain 5 defective and 15 non-defective items, then in a random draw the probability of selecting a defective item (D) is

$$P(D) = \frac{5}{20} = 0.25$$

2.4 Rules of Probability (Two events)

Event	Rule	Explanation
A'	$P(A') = 1 - P(A)$	A does NOT happen(Complement rule)
$A \cap B'$	$P(A \cap B') = P(A) - P(A \cap B)$	Only A happens/ A happens BUT NOT B /Eaxctly A happens
$A \cup B$	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	Either A or B or both/ at least one
$A' \cap B'$	$P(A' \cap B') = P(A \cup B)'$ $= 1 - P(A \cup B)$	Neither A nor B happens

Example: (DC Mont.) If $P(A) = 0.3$, $P(B) = 0.2$, and $P(A \cap B) = 0.1$, determine the following probabilities:

- a. $P(A')$
- b. $P(A \cup B)$
- c. $P(A' \cap B)$
- d. $P(A \cap B')$

- e. $P(A \cup B)'$
- f. $P(A' \cup B)$
- g. $P(A' \cap B')$

Problem 2.1: (W Navidi) A system contains two components, A and B. The system will function so long as either A or B functions. The probability that A functions is 0.95, the probability that B functions is 0.90, and the probability that both function is 0.88. What is the probability that the system functions?

Problem 2.2: (W Navidi) Let S be the event that a randomly selected college student has taken a statistics course, and let C be the event that the same student has taken a chemistry course. Suppose $P(S) = 0.4$, $P(C) = 0.3$, and $P(S \cap C) = 0.2$.

- a. Find the probability that a student has taken statistics, chemistry, or both.
- b. Find the probability that a student has taken neither statistics nor chemistry.
- c. Find the probability that a student has taken statistics but not chemistry.

Problem 2.3: (W Navidi) Let V be the event that a computer contains a virus, and let W be the event that a computer contains a worm. Suppose $P(V) = 0.15$, $P(W) = 0.05$, and $P(V \cup W) = 0.17$.

- a. Find the probability that the computer contains both a virus and a worm.
- b. Find the probability that the computer contains neither a virus nor a worm.
- c. Find the probability that the computer contains a virus but not a worm.

Problem 2.4: (W Navidi) Inspector A visually inspected 1000 ceramic bowls for surface flaws and found flaws in 37 of them. Inspector B inspected the same bowls and found flaws in 43 of them. A total of 948 bowls were found to be flawless by both inspectors. One of the 1000 bowls is selected at random.

- a. Find the probability that a flaw was found in this bowl by at least one of the two inspectors.
- b. Find the probability that flaws were found in this bowl by both inspectors.
- c. Find the probability that a flaw was found by inspector A but not by inspector B.

2.5 Conditional Probability, Multiplication rule and Independence

If A is a prior event (happens first) and B is another event happens after A then the **conditional probability of B given A** is defined as:

$$P(B|A) = \frac{P(A \cap B)}{P(A)} ; P(A) > 0$$

It implies that,

$$P(A \cap B) = P(A) \cdot P(B|A) \quad [\text{Multiplication rule}]$$

Note: Use of these rules depends on the situation (on given information).

Problem 2.5(Walpole) The probability that an automobile being filled with gasoline also needs an oil change is 0.25; the probability that it needs a new oil filter is 0.40; and the probability that both the oil and the filter need changing is 0.14.

- If the oil has to be changed, what is the probability that a new oil filter is needed?
- If a new oil filter is needed, what is the probability that the oil has to be changed?

Independence

If A_1 and A_2 are independent then,

$$P(A_1 \cap A_2) = P(A_1) \cdot P(A_2)$$

For multiple independent events say $A_1, A_2, \dots, A_k$,

$$P(A_1 \cap A_2 \cap \dots \cap A_k) = P(A_1) \cdot P(A_2) \cdots P(A_k)$$

Corrolary: If A_1 and A_2 are independent then,

$$P(A'_1 \cap A'_2) = P(A'_1) \cdot P(A'_2)$$

Problem 2.6(DC Mont.): The probability that a lab specimen contains high levels of contamination is 0.10. Three samples are checked, and the samples are independent.

- What is the probability that none contain high levels of contamination?
- What is the probability that exactly one contains high levels of contamination?
- What is the probability that at least one contains high levels of contamination?

Problem 2.7(Walpole) : A town has two fire engines operating independently. The probability that a specific engine is available when needed is 0.96.

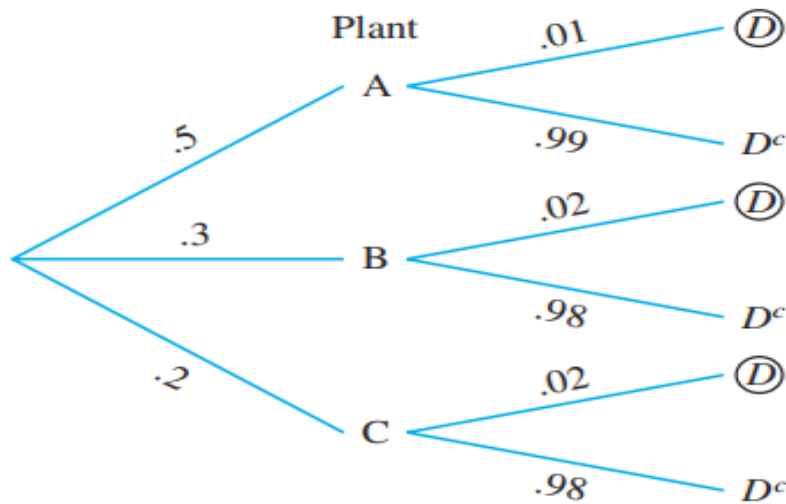
- What is the probability that neither is available when needed?
- What is the probability that a fire engine is available when needed?

2.6 Probability Tree, Total Probability Rule and Bayes' Rule

Given,

Prior probability	Conditional probability
$P(A)=0.5$	$P(D A)=0.01$
$P(B)=0.3$	$P(D B)=0.02$
$P(C)=0.2$	$P(D C)=0.02$

These probabilities can be presented in following tree diagram:



Total probability rule:

$$P(D) = P(D \cap A) + P(D \cap B) + P(D \cap C)$$

$$\text{Or, } P(D) = P(A) \cdot P(D|A) + P(B) \cdot P(D|B) + P(C) \cdot P(D|C)$$

$$\text{Or, } P(D) = (0.5)(0.01) + (0.3)(0.02) + (0.2)(0.02) = 0.015$$

We can also compute $P(A|D)$ as follows:

$$P(A|D) = \frac{P(A \cap D)}{P(D)} = \frac{(0.5)(0.01)}{0.015} = 0.3333 \approx 33.33\%$$

This is known as Bayes' Theorem or rule

Problem 2.8: The panels for the Pulsar 32-inch widescreen LCD HDTVs are manufactured in three locations and then shipped to the main plant of Vista Vision for final assembly. Plants A, B, and C supply 50%, 30%, and 20%, respectively, of the panels used by the company. The quality-control department of the company has determined that 1% of the panels produced by plant A are defective, whereas 2% of the panels produced by plants B and C are defective.

a) What is the probability that a randomly selected Pulsar 32-inch HDTV will have a defective panel? (**Ans:** 0.015)

b) If the selected Pulsar 32-inch HDTV is found defective, what is the probability that the component selected was produced by Plant C? (**Ans:** 26.67%)

Problem 2.9 Suppose 2% of cotton fabric rolls and 3% of nylon fabric rolls contain flaws. Of the rolls used by a manufacturer, 70% are cotton and 30% are nylon.

(a) What is the probability that a randomly selected roll used by the manufacturer contains **flaws**?

(b) If a randomly selected roll is found to contain a flaw, what is the probability that the roll is **cotton**?

2.7 Random Variables

A **variable** is said to be **random** if its values are determined by a random experiment. In other word, **random variable** is a numerical description of the outcome of an experiment.

- A random variable often denoted with an uppercase letter (say X)
- The value of a random variable is denoted with a lowercase letter (say x)

Illustration Consider a random experiment of tossing a coin (fair/unfair) 2 times. Then the sample space is

$$S = \{HH, HT, TH, TT\}$$

Now let, $X = \text{number of heads occur}$

From the sample space we can see that X can take following values:

Sample point	x
HH	2
HT	1
TH	1
TT	0

Since the values of X completely determined by the outcomes of the random experiment, so X is a random variable (discrete).

2.7.1 Types of random variable

There are two types of random variables, **discrete** and **continuous**.

A **discrete random variable** can assume only a certain number of separated values. A discrete random variable is usually the result of *counting* something. For example, number of customers arrive, number of calls receive etc.

A **continuous random variable** is one whose values are uncountable or which can take any value in a given interval. Generally a continuous random variable is usually the result of *measuring* something.

Task Which of these variables are discrete and which are continuous random variables?

- a. The number of new accounts established by a salesperson in a year.
- b. The time between customer arrivals to a bank ATM.
- c. The number of customers in Big Nick's barber shop.
- d. The amount of fuel in your car's gas tank.
- e. The number of minorities on a jury.
- f. The outside temperature today.

3 Discrete Probability Distributions

3.1 Probability distribution of Discrete r.v: Probability mass function (PMF)

Suppose X is a discrete random variable. The **probability mass function (PMF)** of X can be denoted as $f(x)$ where

$$f(x) = P(X = x)$$

For each possible outcome x ; $f(x)$ must satisfies:

1.

$$f(x) \geq 0$$

2.

$$\sum_x f(x) = 1$$

The **PMF** $f(x)$ is also called probability distribution of the discrete random variable X .

3.2 Mean, Variance and Standard deviation of discrete r.v

(a) The **expected value of** X the mean of X is denoted by $E(X)$ and defined by:

$$E(X) = \sum_x x \cdot f(x) \quad (3.1)$$

The expected value of X is sometimes called the population mean of X that is $\mu = E(X)$.

(b) The **variance** of X is

$$Var(X) = \sigma^2 = E[(X - \mu)^2] = \sum_x (x - \mu)^2 f(x)$$

Alternative:

$$Var(X) = E(X^2) - \mu^2 \quad (3.2)$$

Where,

$$E(X^2) = \sum_x x^2 \cdot f(x)$$

Problem 3.1: Consider the following probability distribution of discrete r.v X .

x	1	2	3
$f(x)$	0.25	0.40	0.35

Compute $E(X)$, $Var(X)$ and $SD(X)$.

Solution:

Mean or Expected value of X :

$$\mu = E(X) = \sum_{x=1}^3 x \cdot f(x) = 1 \times 0.25 + 2 \times 0.40 + 3 \times 0.35 = 2.1$$

Variance of X :

$$\begin{aligned} \sigma^2 &= E(X^2) - \mu^2 = \sum_{x=0}^3 x^2 \cdot f(x) - \mu^2 \\ &= [1^2 \times 0.25 + 2^2 \times 0.40 + 3^2 \times 0.35] - (2.1)^2 = 5 - 4.41 = 0.59 \end{aligned}$$

Standard deviation of X :

$$SD(X) = \sigma = \sqrt{0.59} = 0.7681$$

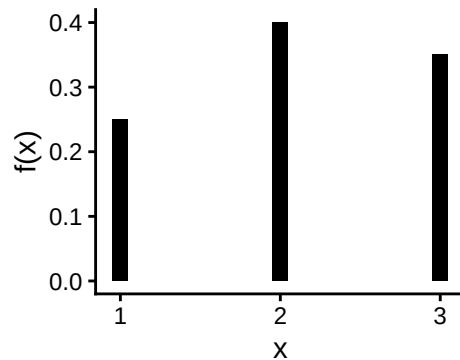


Figure 3.1: Probability histogram of of X (Example 3.1)

Note: $E(X)$ measures the **center** of the distribution and $SD(X)$ measures the **spread** of the distribution around the mean.

3.3 Properties of $E(.)$ and $Var(.)$

If a and b are constants, then

- a) $E(b) = b$
- b) $E(aX + b) = aE(X) + b$
- c) $Var(b) = 0$

d) $Var(aX + b) = a^2 Var(X)$

Exercises 3.1 :

- 1) Let the random variable X represent the number of defective parts for a machine when 3 parts are sampled from a production line and tested. The following is the probability distribution of X .

x	0	1	2	3
$f(x)$	0.51	0.38	0.10	0.01

Calculate average and standard deviation of number of defective parts.

2. A chemical supply company ships a certain solvent in 10-gallon drums. Let X represent the number of drums ordered by a randomly chosen customer. Assume X has the following probability mass function:

x	1	2	3	4	5
$f(x)$	0.4	0.2	0.2	0.1	0.1

- Find** the mean number of drums ordered.
- Find** the variance of the number of drums ordered.
- Find** the standard deviation of the number of drums ordered.
- Let Y be the number of gallons ordered. Find the probability mass function of Y .
- Find** the mean number of gallons ordered.
- Find** the variance of the number of gallons ordered.
- Find** the standard deviation of the number of gallons ordered.

3.4 Binomial distribution

Let X be the number of “successes” in n independent **Bernoulli trial** (a trial which has TWO outcomes-*success or failure*).

Assume $Pr(\text{success}) = p$ (*say*) remains constant from trial to trial. Then the PMF of X is:

PMF:

$$P(X = x) = f(x) = \binom{n}{x} p^x (1 - p)^{n-x} \quad ; \quad x = 0, 1, \dots, n$$

We often use $q = 1 - p$.

With,

- $\sum_{x=0}^n f(x) = 1$
- **Mean:** $\mu = E(X) = np$
- **Variance:** $\sigma^2 = npq = np(1 - p)$

Example: For $n = 3$ and $p = 0.60$ the binomial probabilities are:

$$P(X = 0) = f(0) = \binom{3}{0}(0.60)^0(0.40)^{3-0} = 0.064$$

$$P(X = 1) = f(1) = \binom{3}{1}(0.60)^1(0.40)^{3-1} = 0.288$$

$$P(X = 2) = f(2) = \binom{3}{2}(0.60)^2(0.40)^{3-2} = 0.432$$

$$P(X = 3) = f(3) = \binom{3}{3}(0.60)^3(0.40)^{3-3} = 0.216$$

AND,

$$\text{Mean, } \mu = E(X) = np = 3 \times 0.6 = 1.8$$

$$\text{Variance, } \sigma^2 = npq = 3 \times 0.60 \times 0.40 = 0.72$$

$$SD(X), \sigma = \sqrt{0.72} \approx 0.85$$

Problem 3.2: Let $X \sim \text{Bin}(8, 0.4)$. **Find**

- $P(X = 2)$
- $P(X = 4)$
- $P(X < 2)$
- $P(X > 6)$
- μ_X
- σ_X^2

Problem 3.3: A large industrial firm allows a discount on any invoice that is paid within 30 days. Of all invoices, 10% receive the discount. In a company audit, 12 invoices are sampled at random.

- What is the probability that fewer than 4 of the 12 sampled invoices receive the discount?
- What is the probability that more than 1 of the 12 sampled invoices receives a discount?

Problem 3.4: Of all the weld failures in a certain assembly, 85% of them occur in the weld metal itself, and the remaining 15% occur in the base metal. A sample of 20 weld failures is examined.

- What is the probability that exactly five of them are base metal failures?
- What is the probability that fewer than four of them are base metal failures?
- What is the probability that none of them are base metal failures?

Problem 3.5: Of the items manufactured by a certain process, 20% are defective. Of the defective items, 60% can be repaired.

- Find** the probability that a randomly chosen item is defective and cannot be repaired.

(b). **Find** the probability that exactly 2 of 20 randomly chosen items are defective and cannot be repaired.

i Distribution of sample proportion ($\hat{p}$)

In many situations the success probability p associated with a certain Bernoulli trial remains unknown, and we have to estimate its value. A simple way to do this is to conduct n independent trials and count the number X of successes. To estimate the success probability p we compute the sample proportion $\hat{p}$.

$$\hat{p} = \frac{\text{number of successes}}{\text{number of trials}} = \frac{X}{n}$$

So,

- $E(\hat{p}) = p$
- $Var(\hat{p}) = \frac{p(1-p)}{n}$
- $SD(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$

We will use this concept in upcoming Chapter of **Statistical Quality Control (SQC)** for control charts of **Fraction-Defective**.

3.5 Poisson distribution

Suppose X be the number occurrences in a given **interval** and

μ is the expected value or mean number of occurrences/rate of occurrences in an interval. Then PMF of X is:

PMF:

$$P(X = x) = \frac{e^{-\mu} \mu^x}{x!} \quad ; \quad x = 0, 1, \dots, \infty$$

With,

- $\sum_{x=0}^{\infty} f(x) = 1$
- Mean, $E(X) = \mu$
- Variance, $Var(X) = \sigma^2 = \mu$

We write, $X \sim Pois(\mu)$

Problem 3.6: Let $X \sim Pois(4)$. Find

- a. $P(X = 1)$
- b. $P(X = 0)$
- c. $P(X < 2)$

d. $P(X > 1)$

e. μ_X

f. σ_X

Problem 3.7: A local drugstore owner knows that, on average, 100 people enter his store each hour.

(a) **Find** the probability that in a given 3-minute period nobody enters the store.

(b) **Find** the probability that in a given 5-minute period more than 2 people enter the store.

Problem 3.8: While inspecting a fabric in a continuous roll, the mean number of defects per meter is 0.25. **Determine** the probability that 10 m length of fabric will contain (a) 0, (b) 1, (c) 2, (d) less than or equal to 2 defects.

Problem 3.9: A roll of fabric contains on an average 5 defects scattered at random in 100 m^2 . Pieces of fabric, 5 m by 2 m, are cut from the roll. Determine the probability that

(a) a piece contains no defects;

(b) a piece will contain more than one defects;

(c) a piece will contain fewer than two defects;

(d) five pieces selected at random will all be free from defects.

4 Continuous r.v and probability density function

4.1 Definition

A continuous r.v X must have a probability density function (PDF) $f(x)$ such that

- 1) $f(x) \geq 0$ [**Non-negativity**]
- 2) $\int_{x \in \mathbb{R}} f(x) dx = 1$ [**Total AREA under the curve** $f(x)$ always 1]
- 3) $P(a < X < b) = \int_a^b f(x) dx$

4.2 Illustration with an example

Given $f(x) = \frac{1}{2}x$; $0 \leq x \leq 2$

- a) Show/plot the graph of $f(x)$.
- b) Is $f(x)$ a PDF?
- c) Find $P(X < 1.0)$.
- d) Find $P(X = 1.0)$

i Note

We always remember that **Probability in an interval of X is actually the AREA under the pdf $f(x)$.**

Problem 4.1 A random variable has the following density function.

$$f(x) = 1 - 0.5x \quad ; \quad 0 < x < 2$$

- a) Graph the density function.
- b) Verify that $f(x)$ is a density function.
- c) Find $P(X > 1)$.
- d) Find $P(X < 0.5)$.
- e) Find $P(X = 1.5)$.

N.B: $P(X = a) = 0$ as well as $P(X = b) = 0$. So, $P(X \leq a)$ is same as $P(X < a)$.

4.3 CDF of continuous r.v X

By definition, CDF,

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x)dx$$

Therefore,

- $f(x) = \frac{d}{dx}F(x)$.
- $P(a < X < b) = F(b) - F(a)$.

Problem 4.2 (Walpole et al. 2017) Consider the probability density function (PDF):

$$f(x) = k\sqrt{x} \ ; \ 0 < x < 1$$

- (a) Evaluate k .
- (b) Find $F(x)$ and use it to evaluate $P(0.3 < X < 0.6)$.

Solution:

- (a) Since $f(x)$ is a PDF so

$$\int_{-\infty}^{+\infty} f(x)dx = 1$$

$$\Rightarrow \int_0^1 k\sqrt{x}dx = 1$$

$$\Rightarrow k \int_0^1 x^{\frac{1}{2}} dx = 1$$

$$\Rightarrow k \left[\frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} \right]_0^1 = 1$$

$$\Rightarrow k \cdot \frac{2}{3} \left[x^{\frac{3}{2}} \right]_0^1 = 1$$

$$\Rightarrow k \cdot \frac{2}{3} [1^{3/2} - 0^{3/2}] = 1$$

$$\Rightarrow k \cdot \frac{2}{3} \cdot 1 = 1$$

$$\therefore k = \frac{3}{2}$$

So, $f(x) = \frac{3}{2}\sqrt{x}$ $0 < x < 1$

(b) By definition

$$\begin{aligned} F(x) &= P(X \leq x) \\ &= \int_0^x f(x) \, dx = \int_0^x \frac{3}{2}x^{\frac{1}{2}} dx \\ &= \frac{3}{2} \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^x \\ &= x^{\frac{3}{2}} \end{aligned}$$

$$\therefore F(x) = x^{\frac{3}{2}} \quad ; \quad 0 < x < 1$$

Now,

$$P(0.3 < X < 0.6) = F(0.6) - F(0.3) = (0.6)^{\frac{3}{2}} - (0.3)^{\frac{3}{2}} = 0.3005$$

Problem 4.3 (Walpole et al. 2017) Suppose that the error in the reaction temperature, in $^{\circ}C$, for a controlled laboratory experiment is a continuous random variable X having the probability density function :

$$f(x) = \frac{1}{3}x^2 \quad ; \quad -1 < x < 2$$

(a) **Verify** that $f(x)$ is a PDF.

(b) **Find** $F(x)$ and then **evaluate** $P(0 < X \leq 1)$.

Solution: *TRY yourself.*

4.4 Expectation and variance of continuous r.v

If X is a continuous r.v with PDF $f(x)$ then

Expected value of X is

$$\mu = E(X) = \int_{x \in \mathbb{R}} x \cdot f(x) dx$$

Variance of X is

$$Var(X) = E(X^2) - \mu^2 = \int_{x \in \mathbb{R}} x^2 \cdot f(x) dx - \mu^2$$

4.5 Uniform distribution

A continuous r.v X is said to be uniform r.v ranges between a to b if it has the following PDF

$$f(x) = \frac{1}{b-a} \quad ; \quad a < x < b \quad (4.1)$$

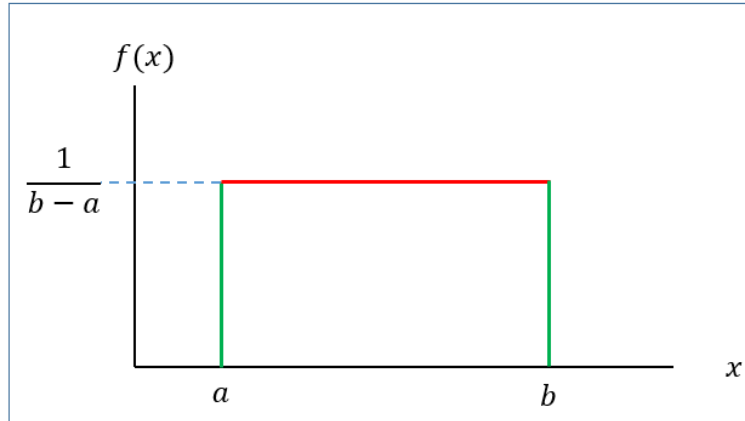


Figure 4.1: Graph of $f(x)$

with

Mean: $\mu = E(X) = \frac{a+b}{2}$

Variance: $\sigma^2 = \frac{(b-a)^2}{12}$

We write, $X \sim U(a, b)$

4.5.1 Finding probability for uniform r.v

If $X \sim U(a, b)$ then the $P(x_1 < X < x_2)$ is actually the **area of the shaded rectangle**.

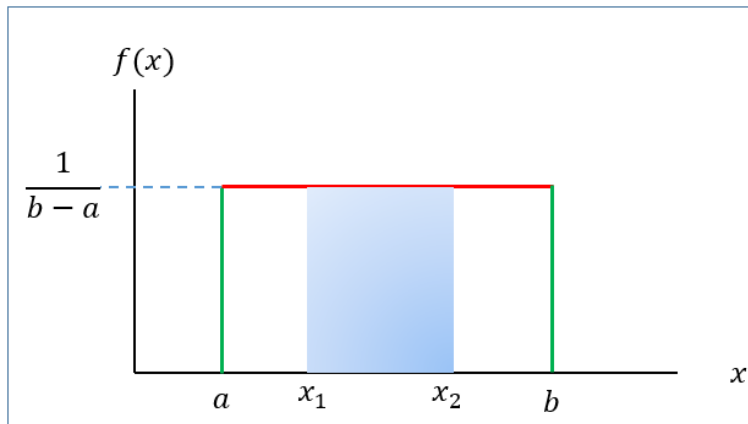


Figure 4.2: Computing area for an interval of Uniform distribution

That is,

$$P(x_1 < X < x_2) = \text{Base} \times \text{Height} = (x_2 - x_1) \times \frac{1}{b - a}$$

Problem 4.4 When a motorist stops at a red light at a certain intersection, the waiting time for the light to turn green, in seconds, is uniformly distributed on the interval $(0, 30)$.

- (a) Find the probability that the waiting time is between 10 and 15 seconds.
- (b) Find the mean and variance of the waiting time.

4.6 Normal or Gaussian r.v

The most important probability distribution for describing a continuous random variable is the **normal probability distribution**. The normal distribution has been used in a wide variety of practical applications in which the random variables are heights and weights of people, test scores, scientific measurements, amounts of rainfall, and other similar values.

4.6.1 Definition

A continuous r.v X is said to be normal r.v if it has the following **PDF**:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} ; -\infty < x < \infty \quad (4.2)$$

The graph of $f(x)$ is called **normal curve** (Figure 4.3).

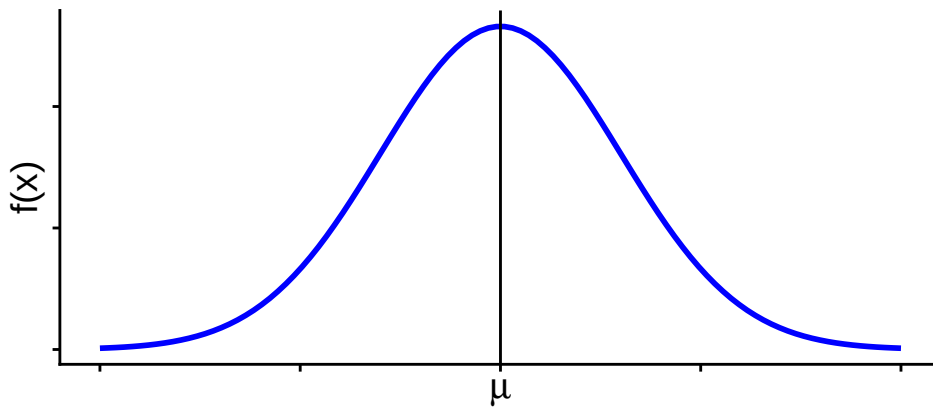


Figure 4.3: Normal Curve

Mean: $E(X) = \mu$

Variance: $Var(X) = \sigma^2$

We write: $X \sim N(\mu, \sigma^2)$

Properties of normal distribution

- The **total area** under the normal curve $f(x)$ is 1 that is

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

- Normal distribution is symmetric about mean, μ
- Mean, median and mode is identical in normal distribution that is $Mean = Median = Mode = \mu$
- Almost 99% observations of X lie within **3 standard deviation of mean** that is

$$P(\mu - 3\sigma < X < \mu + 3\sigma) \approx 0.99$$

- Almost 95% observations of X lie within **2 standard deviation of mean** that is

$$P(\mu - 2\sigma < X < \mu + 2\sigma) \approx 0.95$$

- Almost 68% observations of X lie within **1 standard deviation of mean** that is

$$P(\mu - \sigma < X < \mu + \sigma) \approx 0.68$$

4.6.2 Standard normal r.v

Suppose $X \sim N(\mu, \sigma^2)$. Then the variable $Z = \frac{X-\mu}{\sigma}$ is said to be **standard normal variable** with **PDF**

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} ; -\infty < z < \infty \quad (4.3)$$

Mean: $E(Z) = 0$

Variance: $Var(Z) = 1$

We write: $Z \sim N(0, 1)$

4.6.3 Computing probability(area) under standard normal curve

To compute area (probability) under the standard normal curve for a given interval of z we use **standard Normal Distribution table** which provides cumulative probabilities.

RULE-I: Suppose we want to find $P(Z < 1.25)$.

From **TABLE A.2** in Navidi (2011) we have

$$P(Z < 1.25) = 0.8944$$

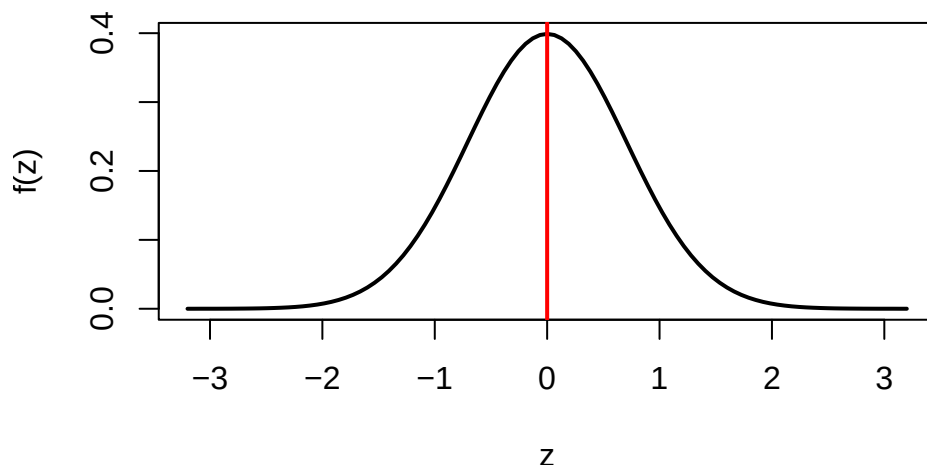
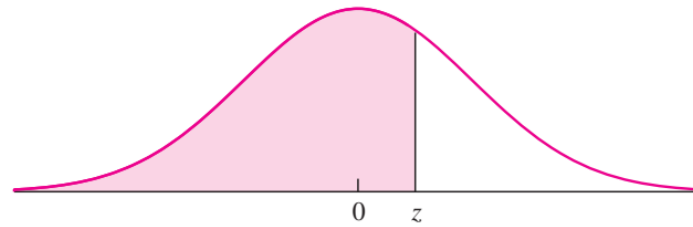


Figure 4.4: Standard normal curve

TABLE A.2 Cumulative normal distribution (continued)



<i>z</i>	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986

The probability $P(Z < 1.25)$ is shown in Figure 4.5.

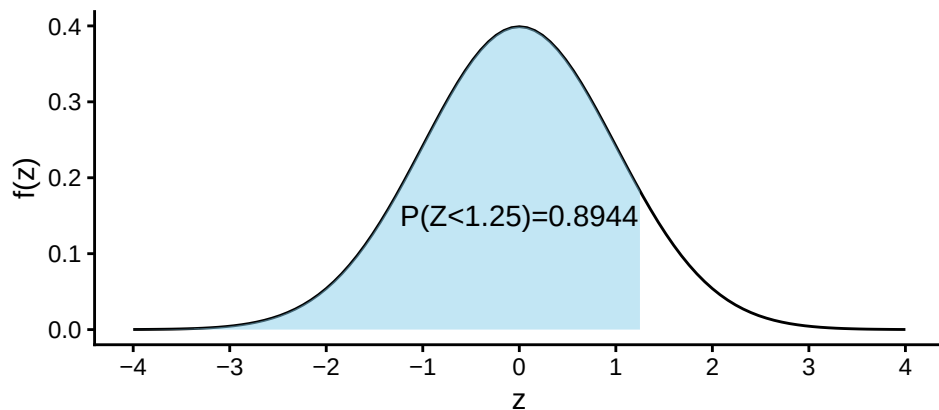


Figure 4.5: Area under standard normal curve for $Z < 1.25$

RULE-II: Now we find $P(Z > 1.36)$

So, due to symmetry we can write $P(Z > 1.36) = P(Z < -1.36) = 0.0869$

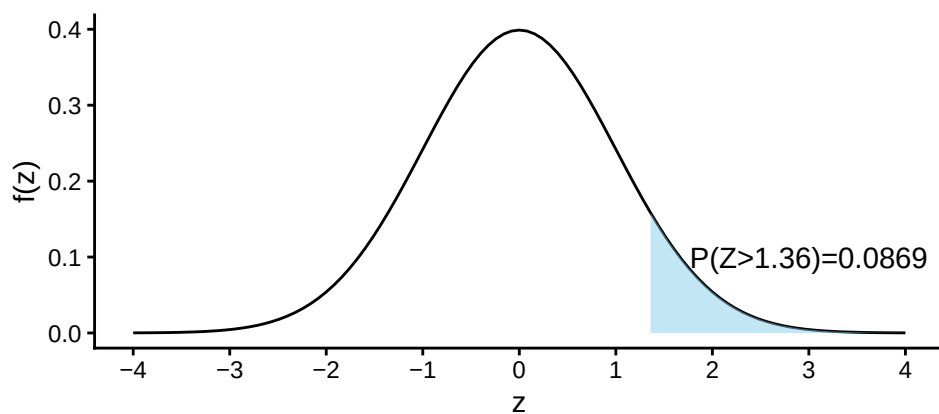


Figure 4.6: Area under standard normal curve for $Z > 1.36$

RULE-III: Let us evaluate $P(-1.96 < Z < 2.58)$.

We can write

$$\begin{aligned}
 &= P(-1.96 < Z < 2.58) \\
 &= P(Z < 2.58) - P(Z < -1.96) \\
 &= 0.9951 - 0.0250 = 0.9701
 \end{aligned}$$

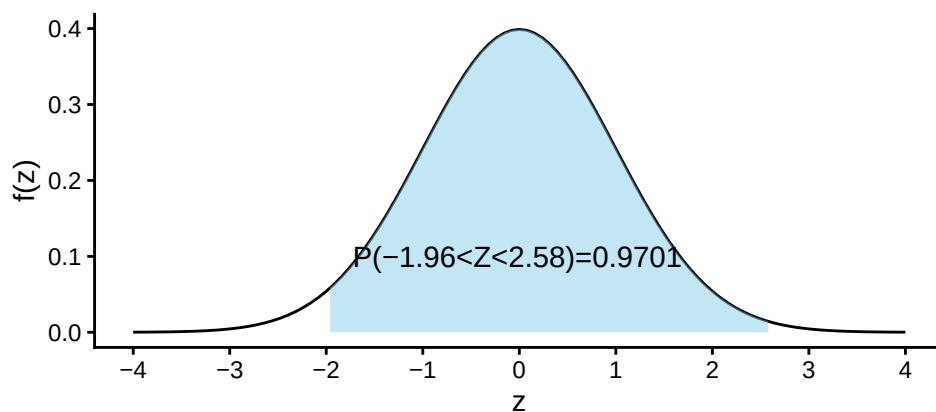


Figure 4.7: Area under standard normal curve for $-1.96 < Z < 2.58$

4.6.4 Finding quantiles (percentiles, quartiles, deciles etc.) of Z

What is the 90th percentile of Z ? To answer this question, let k is the 90th percentile of Z . So we can write

$$P(Z < k) = 0.90 \quad \dots (1)$$

From **TABLE A.2** in Navidi (2011) we have

$$P(Z < 1.28) = 0.90 \quad \dots (2)$$

Comparing eq.(1) with eq.(2) we have $k = 1.28$. So the 90th percentile of Z is 1.28.

Problem 1 Find c such that $P(Z > c) = 0.05$.

Problem 2 Find c such that $P(-c < Z < c) = 0.95$.

4.6.5 Computing probability(area) under normal curve:

Problem 4.5 Suppose $X \sim N(30, 5^2)$. Then find the following:

- $P(X < 22)$
- $P(X > 44)$
- $P(20 < X < 35)$
- If $P(X < x) = 0.25$ then find the value of x .

Solution:

Here, $\mu = 30$ and $\sigma = 5$

$$\text{a) } P(X < 22) = P\left(\frac{X - \mu}{\sigma} < \frac{22 - 30}{5}\right) = P(Z < -1.60) = 0.0548.$$

$$\begin{aligned} \text{b) } P(X > 44) &= P\left(\frac{X-\mu}{\sigma} > \frac{44-30}{5}\right) \\ &= P(Z > 2.80) = P(Z < -2.80) = 0.0026 \end{aligned}$$

$$\begin{aligned} \text{c) } P(20 < X < 35) &= P\left(\frac{20-30}{5} < \frac{X-\mu}{\sigma} < \frac{35-30}{5}\right) \\ &= P(-2 < Z < 1) = P(Z < 1) - P(Z < -2) \\ &= 0.8413 - 0.0228 = 0.8185 \end{aligned}$$

d) To find the value of x we proceed this way.

$$P(X < x) = 0.25$$

$$\Rightarrow P\left(\frac{X-\mu}{\sigma} < \frac{x-30}{5}\right) = 0.25$$

$$\Rightarrow P\left(Z < \frac{x-30}{5}\right) = 0.25 \quad \dots (1)$$

From TABLE (Appendix B) we have

$$P(Z < -0.67) = 0.25 \quad \dots (2)$$

Comparing (1) with (2) we can write

$$\frac{x-30}{5} = -0.67$$

$$\Rightarrow x = 30 + (-0.67) \times 5$$

$$\therefore x = 26.65$$

i Note

If $P(X < x) = p$ and
 $P(Z < z) = p$ then

$$x = \mu + z\sigma$$

4.6.6 Applications of the Normal Distribution (Walpole et al. 2017)

Example 6.7: A certain type of storage battery lasts, on average, 3.0 years with a standard deviation of 0.5 year. Assuming that battery life is normally distributed, find the probability that a given battery will last less than 2.3 years.

Example 6.8 : An electrical firm manufactures light bulbs that have a life, before burn-out, that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a bulb burns between 778 and 834 hours.

***Example 6.9 :** In an industrial process, the diameter of a ball bearing is an important measurement. The buyer sets specifications for the diameter to be 3.0 ± 0.01 cm. The implication is that no part falling outside these specifications will be accepted. It is known that in the process the diameter of a ball bearing has a normal distribution with mean $\mu = 3.0$ and standard deviation $\sigma = 0.005$. On average, how many manufactured ball bearings will be scrapped?

Example 6.10 : Gauges are used to reject all components for which a certain dimension is not within the specification $1.50 \pm d$. It is known that this measurement is normally distributed with a mean of 1.50 and a standard deviation of 0.2. Determine the value d such that the specifications “cover” 95% of the measurements.

Solution:

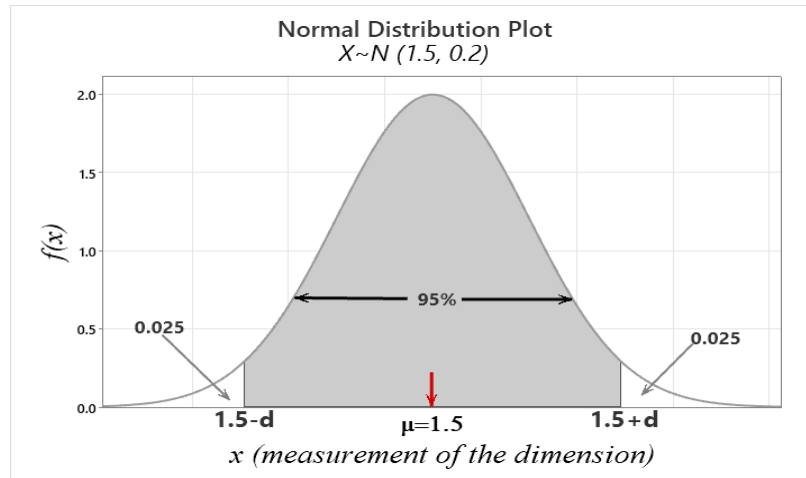
Let, $X = \text{measurement of certain dimension}$

Given, $X \sim N(1.5, 0.2)$

According to question,

$$P(1.5 - d < X < 1.5 + d) = 0.95$$

$$\text{So, } P(X < 1.5 - d) + P(X > 1.5 + d) = 0.05$$



$$\text{So, } P(X < 1.5 - d) = 0.025 \quad \dots (1)$$

$$\Rightarrow P\left(\frac{X - \mu}{\sigma} < \frac{1.5 - d - 1.5}{0.2}\right) = 0.025$$

$$\Rightarrow P\left(Z < \frac{-d}{0.2}\right) = 0.025 \quad \dots (1)$$

From TABLE A.2 we have

$$P(Z < -1.96) = 0.025 \quad \dots (2)$$

Comparing eq.(1) with eq. (2) we have

$$\frac{-d}{0.2} = -1.96$$

$$\implies -d = -0.392$$

$$\therefore d = 0.392$$

Alternative:

$$P(X < 1.5 - d) = 0.025 \quad \dots (1)$$

$$\text{But, } P(Z < -1.96) = 0.025 \quad \dots (2)$$

Hence, $1.5 - d = \mu + z\sigma$; where $z = -1.96$

$$\implies 1.5 - d = 1.5 - 1.96 \times 0.2$$

$$\therefore d = 0.392$$

***Exercise 6.11 :** A soft-drink machine is regulated so that it discharges an average of 200 milliliters per cup. If the amount of drink is normally distributed with a standard deviation equal to 15 milliliters,

- what fraction of the cups will contain more than 224 milliliters?
- what is the probability that a cup contains between 191 and 209 milliliters?
- how many cups will probably overflow if 230-milliliter cups are used for the next 1000 drinks?
- below what value do we get the smallest 25% of the drinks?

***Exercise 6.14** The finished inside diameter of a piston ring is normally distributed with a mean of 10 centimeters and a standard deviation of 0.03 centimeter.

- What proportion of rings will have inside diameters exceeding 10.075 centimeters?
- What is the probability that a piston ring will have an inside diameter between 9.97 and 10.03 centimeters?
- Below what value of inside diameter will 15% of the piston rings fall?

Exercise 6.17 : The average life of a certain type of small motor is 10 years with a standard deviation of 2 years. The manufacturer replaces free all motors that fail while under guarantee. If she is willing to replace only 3% of the motors that fail, how long a guarantee should be offered? Assume that the lifetime of a motor follows a normal distribution.

4.7 Some important derived continuous distribution

4.7.1 Chi-square (χ^2) distribution

The continuous random variable X has a chi-squared distribution, with ν degrees of freedom, if its density function is given by

$$f(x) = \frac{1}{2^{\nu/2}\Gamma(\nu/2)} x^{\nu/2-1} e^{-x/2} \quad ; \quad x > 0$$

Where ν is a positive integer.

The mean and variance of the chi-squared distribution are

$$\mu = \nu \text{ and } \sigma^2 = 2\nu$$

4.7.2 t -distribution

4.7.3 F - distribution

5 Estimation and Sampling Distribution

5.1 Introduction

Estimation refers to finding the value of an unknown *population parameter* based on the sample selected from that population. Estimations are of two types- (i) Point estimation (ii) Interval estimation. First we discuss about point estimation, then sampling distribution of point estimators which will be needed to understand interval estimation.

5.1.1 Point Estimation

Point estimation involves **sample statistic**(*any function of the observations in a random sample*) , a formula or function of sample statistics (*point estimator*) and **estimate** (*the numerical value of the point estimator*).

For example, suppose we draw a random sample say $x_1, x_2, \dots, x_n$ from a population regarding a variable X . Here

- Parameter of interest: Population mean, μ
- Point estimator of μ : Sample mean $\bar{X} = \frac{\sum x}{n}$

If the we have a sample of size $n = 4$ and the values are: $x_1 = 25, x_2 = 30, x_3 = 29, x_4 = 31$, then the point estimate of μ is:

$$\bar{x} = \frac{25 + 30 + 29 + 31}{4} = 28.75$$

Another important population parameter is population variance, σ^2 . And its point estimator is $S^2 = \frac{\sum_{i=1}^n (x_i - \bar{X})^2}{n-1}$

Note: Due to sampling variability **point estimators** are random variables.

5.2 Sampling distribution

! Theorem 5.1: Central Limit Theorem

i Theorem 5.2

Let $X_1, X_2, \dots, X_n$ be independent random variables that are all normal with mean μ and standard deviation σ . Let

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{and} \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Then the random variable $T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$ has a t -distribution with $v = n - 1$ degrees of freedom.

5.3 Interval estimation

5.3.1 Interval estimation for μ (when σ is known)

5.3.2 Interval estimation for μ (when σ is Unknown)

5.3.3 Interval estimation for μ in case of Large Sample

6 Significance test: Statistical Hypothesis Testing

6.1 Definition Statistical Hypothesis

A statistical hypothesis is a *statement* about the *parameters* of one or more populations.

Example 1: A manufacturer claims that the mean life of a smartphone is more than 1.5 years.

Example 2: A local courier service claims that they deliver a ordered product within 30 minutes on average.

Example 3: A sports drink maker claims that the mean calorie content of its beverages is 72 calories per serving.

6.2 Types of hypothesis

Statistical hypothesis are stated in two forms- (i) Null hypothesis (H_0) and (ii) Alternative hypothesis (H_1).

Both null and alternative hypothesis are the written about the parameter of interest based on the claim.

- We will always state the null hypothesis as an **equality claim**.
- However, when the alternative hypothesis is stated with the “<” sign, the implicit claim in the null hypothesis can be taken as “ ” or “=” sign.
- When the alternative hypothesis is stated with the “>” sign, the implicit claim in the null hypothesis can be taken as “ ” or “=” sign.

6.3 Developing hypotheses

To develop or state null and alternative hypothesis, at first we have to clearly identify the “**claim**” about population parameter. Now we will see some examples.

Example 1: A manufacturer claims that the mean life of a smartphone is more than 1.5 years.

Hypothesis:

$$H_0 : \mu = 1.5$$

$$H_1 : \mu > 1.5 \text{ (claim)}$$

Example 2: A local courier service claims that they deliver a ordered product within 30 minutes on average.

Hypothesis:

$$H_0 : \mu = 30$$

$$H_1 : \mu < 30 \text{ (claim)}$$

Example 3: A sports drink maker claims that the mean calorie content of its beverages is 72 calories per serving.

Hypothesis:

$$H_0 : \mu = 72 \text{ (claim)}$$

$$H_1 : \mu \neq 72$$

6.4 Types of test based on alternative hypothesis H_1

- $H_1 : \mu < \mu_0$ (Lower tailed)
- $H_1 : \mu > \mu_0$ (Upper tailed)
- $H_1 : \mu \neq \mu_0$ (Two-tailed)

6.5 Types of error in hypothesis test and P-value

While testing a statistical hypothesis concerning population parameter we commit two types of errors.

- **Type I error** occurs when we **reject** a **TRUE** H_0
- **Type II error** occurs when we **FAIL to reject** a **FALSE** H_0
- The **Level of significance** is the probability of committing **Type I error**. It is denoted by α .

$$\alpha = P(\text{Type I error})$$

- The probability of committing a **Type II error**, denoted by β .

$$\beta = P(\text{Type II error})$$

! Note

Type I error is more serious than **Type II error**. Because rejecting a TRUE statement is more devastating than FAIL to reject a FALSE statement. So, we always try to keep our probability of Type I error as small as possible (1% or at most 5%). For more detail see (**keller2014?**).

- **P-value:** If the null hypothesis is true, then a P-value (or probability value) of a hypothesis test is the probability of obtaining a sample statistic with a value as extreme or more extreme than the one determined from the sample data.

The smaller the P-value of the test, the more evidence there is to reject the null hypothesis. A very small P-value indicates an unusual event

! P-value Explanation

A hypothesis test functions like a **legal trial**. You start by assuming the **null hypothesis** (H_0) is true, just as a jury assumes *a defendant is innocent until proven guilty*.

The data you collect acts as the evidence. By comparing this evidence to your initial assumption, you calculate a **p-value** (ranging from 0 to 1), which quantifies how well the sample data “fits” with H_0 .

- A **high p-value** suggests the evidence is consistent with the null hypothesis.
- A **low p-value** indicates a sharp disagreement between the data and the null hypothesis.

When the p-value is small enough, it provides “beyond a reasonable doubt” proof that the initial assumption is unlikely. At this point, you **reject the null hypothesis** and shift your belief to the **alternative hypothesis** (H_1).

So, how these hypotheses will be tested?

To test a hypothesis we have to determine

- a **test-statistic**; and
- **Critical/Rejection region** based on the sampling distribution of test-statistic for a given α ;
- If the value of test-statistic **falls in Critical/Rejection region**, then we reject Null (H_0) hypothesis; otherwise not. Or,
- If **P-value** $\leq \alpha$ then we reject Null (H_0) hypothesis.

6.6 Hypothesis testing concerning population mean (μ)

The following two hypotheses tests are used concerning population mean (μ):

1. One sample z-test (with known σ)
2. One sample t-test (with unknown σ)

6.6.1 One sample z-test

When sampling is from a **normally distributed population** or **sample size is sufficiently large** and **the population variance is known**, the test statistic for testing $H_0 : \mu = \mu_0$ at α is

$$z_0 = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Decision (Critical value approach): If calculated z falls in rejection region (CR) , then reject H_0 . Otherwise, do not reject H_0 .

- For lower tailed test, reject H_0 if $z_0 < -z_\alpha$;
- For upper tailed test, reject H_0 if $z_0 > z_\alpha$;
- For two-tailed test, reject H_0 if $z_0 < -z_{\alpha/2}$ or $z_0 > z_{\alpha/2}$.

NB: Since z -distribution is symmetric about 0 so, a shortcut technique in CR/RR approach is:

- Reject H_0 if $|z_0| > |z_c|$; otherwise do not.

Decision (P-value approach)

Table 6.1: P-value for z -statistic

Alternative hypothesis	P-value	Calculator function
$H_1 : \mu < \mu_0$ (Left Tailed)	Area in left tail = $P(Z < z_0)$	$P(z_0)$
$H_1 : \mu > \mu_0$ (Right Tailed)	Area in right tail = $P(Z > z_0)$	$R(z_0)$
$H_1 : \mu \neq \mu_0$ (Two-tailed)	Area in both tails = $2P(Z < - z_0)$ OR, $2P(Z > z_0)$	$2P(- z_0)$ OR, $2R(z_0)$

Problem 6.1: A new concrete mix is being designed to provide adequate compressive strength for concrete blocks. The specification for a particular application calls for the blocks to have a mean compressive strength μ greater than 1350 kPa. A sample of 100 blocks is produced and tested. Their mean compressive strength is 1356 kPa .Assume population standard deviation is 70 kPa (known).

- Reframe/ formulate/** state the hypothesis.
- Compute** appropriate test statistic.
- Compute** P-value.
- Critique/ Justify whether the blocks meet the specification.

Problem 6.2: A fabric manufacturer placed an order to a yarn manufacturer for a large batch of yarn that has strength of 20 cN/tex. He/she will ask for replacement if mean strength is less than 20cN/tex.

Upon delivery, the fabric manufacturer found that the average strength of the yarn is 19.65 cN/tex with 1.2 standard deviation based on 40 tests.

Carry out a statistical test at 0.05 level of confidence to justify whether the fabric manufacturer would ask the yarn manufacturer for replacement of yarn batch.

Problem 6.3: A process that manufactures steel bolts is supposed to be calibrated to produce bolts with a mean length of 5 cm. A sample of 100 bolts has a mean length of 5.02 cm with a standard deviation of 0.06 cm. Let μ be the mean length of bolts manufactured by this process .

- (a) **Evaluate/Estimate** the P-value for testing $H_0 : \mu = 5$ versus $H_1 : \mu \neq 5$.
- (b) **Judge/Justify** whether the process is properly calibrated or not.

Problem 6.4: A certain type of stainless steel powder is supposed to have a mean particle diameter of $\mu = 15\mu m$. A random sample of 87 particles had a mean diameter of $15.2\mu m$, with a standard deviation of $1.8\mu m$. A test is made of $H_0 : \mu = 15$ versus $H_1 : \mu \neq 15$.

- a. **Find** the P-value.
- b. Do you believe it is plausible that the mean diameter is $15\mu m$, or are you convinced that it differs from $15\mu m$? Explain your reasoning.

6.6.2 One-sample t-test

6.7 Two-sample z-test

6.8 Two-sample t-test (Assuming equal variances)

6.9 Variance test: Single population

6.10 Variance test: Two populations

6.11 Test of Normality

6.11.1 Graphical Methods

6.11.2 Numerical methods

7 Regression Analysis

7.1 SLR

! Summary of Simple Linear Regression (SLR)

1) The PRF : $y_i = \alpha + \beta x_i + \epsilon_i$

2) The OLS estimates of α and β are:

$$a = \bar{y} - b\bar{x}$$

$$b = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

respectively.

Hence the **estimated regression line** is:

$$\hat{y}_i = a + bx$$

3) **Point prediction:** For a given value of X say x_g the predicted value of Y is:

$$\hat{y}_g = a + b * (x_g)$$

4) **Goodness of fit for SLR:**

$$R^2 = \frac{b^2(\sum x^2 - n \bar{x}^2)}{\sum y^2 - n \bar{y}^2} ; \quad 0 \leq R^2 \leq 1$$

5) **Mean squared error (MSE): as an estimate of σ_e^2 :**

$$s_e^2 = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - 2}$$

6) **Hypothesis test for β :**

$$H_0 : \beta = c \quad vs.$$

$$H_1 : \beta \neq c$$

Test statistic:

$$t_0 = \frac{b - c}{s.e(b)}$$

where, $s.e(b) = \frac{s_e}{\sqrt{\sum (x - \bar{x})^2}}$

Under H_0 , t_0 follows T -distribution with $n - 2$ df.

Problem 7.1 (Walpole et al. 2017, 399): The following is a portion of a classic data set called the “pilot plot data” in Fitting Equations to Data by Daniel and Wood, published in 1971. The response y is the acid content of material produced by titration, whereas the regressor x is the organic acid content produced by extraction and weighing.

y	76	62	66	58	88	70	37	82	88	43
x	123	55	100	75	159	109	48	138	164	28

7.2 MLR

8 DoE

8.1 Single factor: ANOVA

8.2 Two-factor experiment: Factorial ANOVA

9 Statistical Quality Control

9.1 Control Charts

9.2 Process Capability

10 Summary

In summary, this book is written in a concise format.

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